



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO BALUSWAMY AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

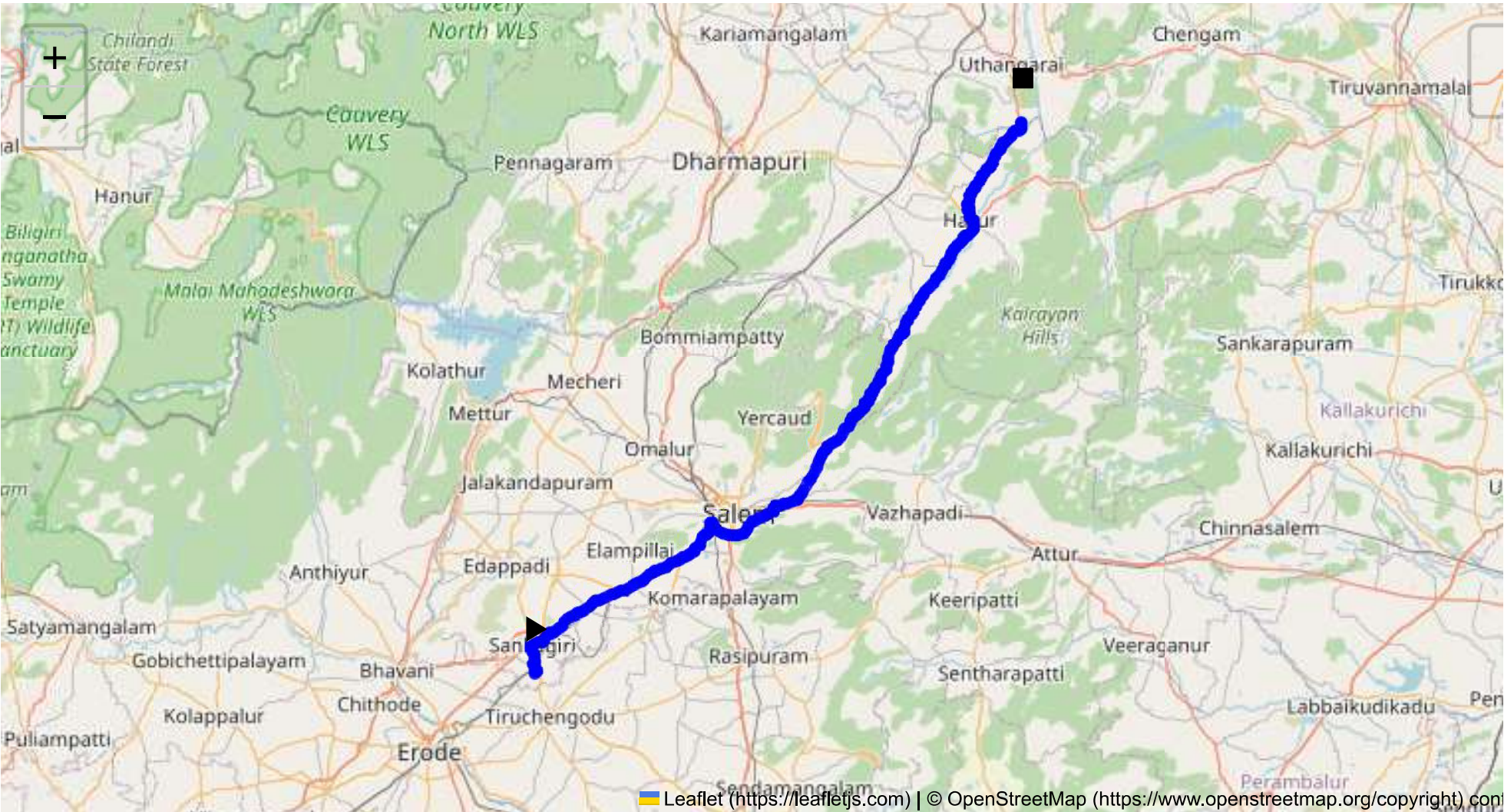
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 124.39 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.186054, 78.550048)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri, Tamil Nadu to Lakkampatti, Tamil Nadu covers approximately 124.39 kilometers, typically taking about 2.93 hours for heavy vehicles, particularly those carrying hazardous materials. The primary roads involved include NH 44, NH 544, and state highways. The journey traverses both rural and semi-urban areas with a mix of straight and winding paths.

Weather Conditions

Tamil Nadu generally experiences a tropical climate. During summer months, temperatures can soar, potentially affecting vehicle performance, especially for loaded trucks. The monsoon season between June and September can bring heavy rains, leading to waterlogging and slippery roads. This may impact visibility and braking efficiency, increasing accident risk.

Traffic Patterns

- **Peak Hours:** Morning (7:30–9:30 AM) and evening (5–7 PM) times may see increased congestion.
- **Congestion Spots:** Urban intersections, toll booths, especially near major towns along NH 44 and NH 544.
- It is advisable to avoid traveling during high-traffic periods to minimize delays.

Road Quality and Infrastructure

- **NH 44 and NH 544:** Generally well-maintained with multiple lanes, though vigilance is required for unexpected potholes or road repairs.
- **State Highways:** These might be narrower with occasional uneven surfaces, requiring careful navigation.

Alternative Routes for Emergencies

In case of severe congestion or blockage:

- Consider diverting through nearby towns via district roads. However, this could increase travel time and complexity due to smaller roads.
- It's crucial to keep updated with local traffic advisories.

Local Regulations for Hazardous Materials Transport

- Ensure compliance with regulations regarding hazardous material signs.
- Adhere to speed limits, especially when transporting dangerous goods.
- Only travel during designated hours and avoid restricted routes for such materials.

Historical Incidents

- Historical data should be reviewed from local and national transport safety agencies, as specific incident rates were not directly accessible, alluding to sporadic truck accidents mostly on narrower state roads.
- Focus should be on ensuring load security and cautious driving to prevent similar incidents.

Environmental Considerations

- No significant environmental sensitive areas are noted along the primary roads. However, rural parts may have protected land or water sources adjacent to the route, so avoid off-road driving near those

areas to prevent contamination.

Communication Coverage

- Major highways like NH 44 and NH 544 have good coverage by major network providers.
- Some rural stretches, especially on state and district roads, might have spotty service. Equip drivers with alternate communication means such as satellite phones for emergencies.

Emergency Response Times

- **Urban areas:** Fast response times, less than 30 minutes typically.
- **Rural stretches:** Slower due to distance from emergency services, potentially exceeding an hour.

Overall Risk Assessment

- **Low to Moderate Risk Level:** The primary routes are well-serviceable but require caution due to potential weather impacts, intermittent traffic congestion, and specific segments with lower infrastructure reliability.
- Drivers should maintain regular communication, follow safety protocols, and be prepared with contingency plans to mitigate identified risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	U-Turn	High	12.1865904, 78.5493678	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53501, 77.95707	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.53601, 77.95856	30 KM/Hr
14	Turn	Medium	11.62403, 78.10893	30 KM/Hr
15	Turn	High	11.64172, 78.11817	15 KM/Hr
16	Turn	Medium	11.64224, 78.11928	30 KM/Hr
17	Turn	High	11.64195, 78.12129	15 KM/Hr
18	Turn	High	11.64120, 78.12168	15 KM/Hr
19	Turn	Medium	11.63948, 78.12081	30 KM/Hr
20	Turn	Medium	11.62861, 78.16526	30 KM/Hr
21	Turn	Medium	11.66388, 78.20672	30 KM/Hr
22	Turn	Medium	11.67129, 78.22978	30 KM/Hr
23	Turn	Medium	11.67116, 78.23102	30 KM/Hr
24	Turn	High	11.67145, 78.23141	15 KM/Hr
25	Turn	High	11.67198, 78.23148	15 KM/Hr
26	Turn	Medium	11.67440, 78.23914	30 KM/Hr
27	Turn	Medium	11.80440, 78.33325	30 KM/Hr
28	Turn	Medium	11.81235, 78.34084	30 KM/Hr
29	Turn	Medium	11.83482, 78.35294	30 KM/Hr
30	Turn	Medium	11.84738, 78.36062	30 KM/Hr
31	Turn	Medium	11.99554, 78.44553	30 KM/Hr
32	Turn	Medium	12.01577, 78.45673	30 KM/Hr
33	Turn	Medium	12.04335, 78.48368	30 KM/Hr
34	Turn	Medium	12.04808, 78.48399	30 KM/Hr
35	Turn	Medium	12.04979, 78.48303	30 KM/Hr
36	Turn	Medium	12.06932, 78.47655	30 KM/Hr
37	Turn	Medium	12.07146, 78.47731	30 KM/Hr
38	Turn	Medium	12.07499, 78.47609	30 KM/Hr
39	Turn	Medium	12.11536, 78.49593	30 KM/Hr
40	Turn	Medium	12.17739, 78.54764	30 KM/Hr
41	Turn	Medium	12.17798, 78.54831	30 KM/Hr
42	Turn	Medium	12.18550, 78.54960	30 KM/Hr
43	Blind Spot	Blind Spot	12.18659, 78.54937	10 KM/Hr
44	Turn	Medium	12.18609, 78.54973	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
26	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
27	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
28	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
29	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
31	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
33	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
34	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
42	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
43	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium
44	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
30	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
32	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
35	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
36	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
37	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
38	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
39	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
40	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
41	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 12.1865904, 78.5493678



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62403, 78.10893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64224, 78.11928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62861, 78.16526



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66388, 78.20672



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67129, 78.22978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67116, 78.23102



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67145, 78.23141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67198, 78.23148



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80440, 78.33325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81235, 78.34084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83482, 78.35294



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84738, 78.36062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99554, 78.44553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01577, 78.45673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04335, 78.48368



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04808, 78.48399



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04979, 78.48303



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.06932, 78.47655



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07146, 78.47731



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07499, 78.47609



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11536, 78.49593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17739, 78.54764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17798, 78.54831



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.18550, 78.54960



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.18659, 78.54937



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.18609, 78.54973

